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Assignment 7

Routing using OSPF (Link State Routing) in Cisco Packet Tracer

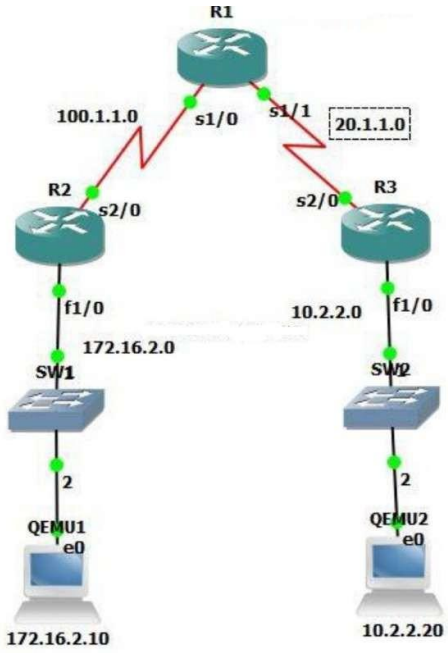
Objective

- To understand the **Link State Routing algorithm (OSPF)**
- To configure and verify **OSPF routing** in a network
- To analyze how **Shortest Path First (SPF)** algorithm works

Problem Statement

Design and implement the given network topology with three routers (R1, R2, R3), two LANs, and PCs.

In addition to configuration, **analyze route selection using OSPF cost and SPF algorithm**.



Network Topology & IP Addressing

- 3 Routers: R1, R2, R3
- 2 LANs
- Serial links between routers
- PCs connected to R2 and R3

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Tasks

Part A: Basic Configuration

1. Create the network topology in Cisco Packet Tracer.
2. Configure IP addresses on all router interfaces and PCs.
3. Verify connectivity between directly connected devices

Part B: OSPF Configuration

1. Enable OSPF on all routers:

router ospf 1

2. Configure router IDs manually (recommended):

router-id 1.1.1.1

3. Advertise networks using:

network <network-address> <wildcard-mask> area 0

4. Ensure all routers are in **Area 0 (Backbone Area)**

Sample Configuration

R1

```
router ospf 1
router-id 1.1.1.1
network 100.1.1.0 0.0.0.255 area 0
network 20.1.1.0 0.0.0.255 area 0
```

R2

```
router ospf 1
router-id 2.2.2.2
network 100.1.1.0 0.0.0.255 area 0
network 172.16.0.0 0.0.255.255 area 0
```

R3

```
router ospf 1
router-id 3.3.3.3
network 20.1.1.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.255 area 0
```

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Part C: Link State (OSPF) Analysis (NEW)

Step 1: Neighbor Discovery

- Use:

show ip ospf neighbor

- Identify OSPF neighbors for each router

Step 2: Link State Database (LSDB)

- Use:

show ip ospf database

- Observe LSAs shared between routers

Step 3: SPF Calculation (Conceptual)**

Students must:

- Draw network graph
- Assign **cost to each link** (default or manual)
- Apply **Shortest Path First (Dijkstra Algorithm)**

◆ **Part D: Verification Commands**

- Check routing table:

show ip route

- Check OSPF:

show ip ospf neighbor
show ip ospf database

◆ **Part E: Simulation & Testing**

1. Ping from PC1 → PC2
2. Use:

traceroute

3. Change bandwidth on one link:

bandwidth 64

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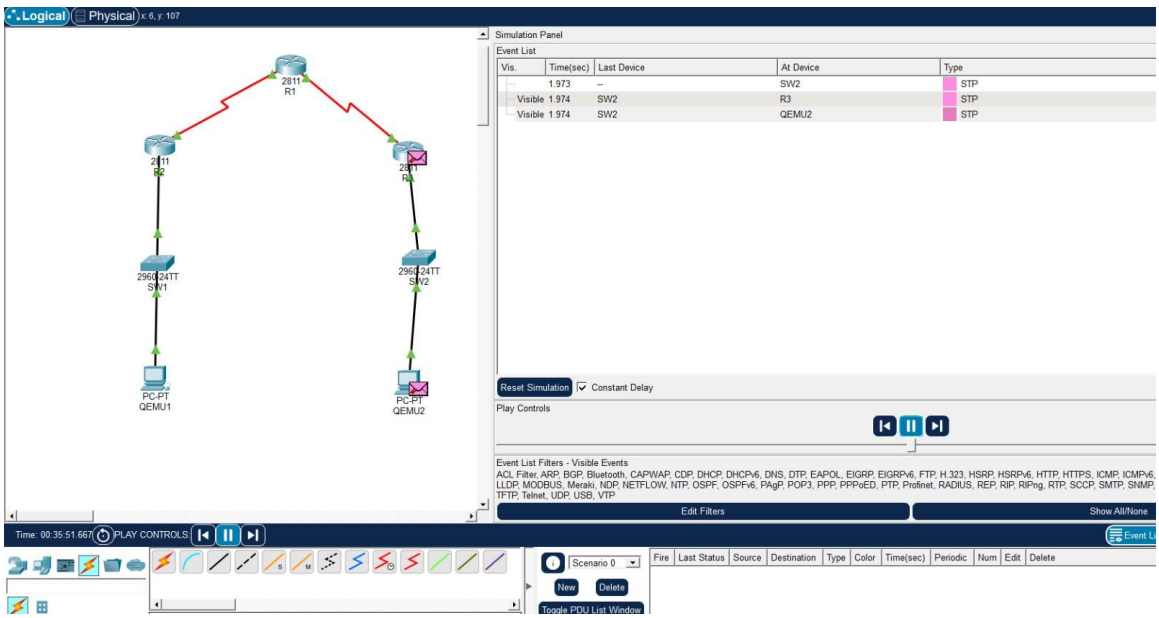
4. Observe route change based on cost

Additional Analytical Task (NEW)

Fill the table:

Router	Destination	Path	Cost
R1	10.2.2.0	R1→R3	?

Solution